

Course- B.Tech.	Type- Core
Course Code- 2025CSET106	Course Name- Discrete Mathematical Structures
Year- I	Semester- Even (2 nd Semester)

Tutorial-6

Tutorial No.	Name	CO1	CO 2	CO3
6	Fuzzy Sets, Set Identities.	✓	--	✓

Objectives

- Students will be able to learn the proofs of classical and fuzzy set theoretic results.
- Students will be able to learn about set identities.

Questions:

- Prove that
 - $\overline{A \cap B} = \overline{A} \cup \overline{B}$.
 - $\overline{A \cup B} = \overline{A} \cap \overline{B}$
- A crisp set A defined over $X = \{1,2,3,5,7\}$ is $A = \{1,3,7\}$. What would be its equivalent fuzzy set?
- A membership function $\mu(x) = x/(1-x) \forall x \in \mathbf{R}$, $\mathbf{R}$ is the set of all real numbers. Check whether $\mu(x)$ can qualify as a member function?
- Define union, intersection, complement, and cardinality of fuzzy set. Let two sets A and B be two fuzzy sets defined on a set $X = \{10,20,30,40,50\}$ whose membership functions are defined as $\mu_A(x) = x/(x + 10)$ and $\mu_B(x) = 2x/(3x+10)$. Then find
 - A and B
 - $A \cap B$, $|A \cap B|$
 - $A \cup B$, $|A \cup B|$
- Let the universal set is $X = \{a,b,c,d\}$. If $A = \{(a,0.2), (b, 0.7), (c, 1), (d, 0.9)\}$, $B = \{(a,0.1), (b, 0.8), (d, 0.2)\}$, then find
 - $A \cap B$, $|A \cap B|$
 - $A \cup B$, $|A \cup B|$
 - A^c , $|A^c|$
 - Support and core of A and B.
- Let A, B and C be sets. Show that $\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}$
- Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i, $A_i = \{1,2,3 \dots i\}$.
- Use set builder notation and logical equivalences to establish the first De Morgan law

$$\overline{A \cap B} = \overline{A} \cup \overline{B}.$$